








Advanced Time Series Analysis and Forecasting

Seminar 0: Refresher and inference for dependent data



Daniel Traian PELE

Bucharest University of Economic Studies
IDA Institute Digital Assets
Blockchain Research Center
AI4EFin Artificial Intelligence for Energy Finance
Romanian Academy, Institute for Economic Forecasting
MSCA Digital Finance

Acronyms used in this material

- **ACF**: Autocorrelation Function
- **AI**: Artificial Intelligence
- **AR**: AutoRegressive (model)
- **ARMA**: AutoRegressive Moving Average (model)
- **BET**: Bucharest Exchange Trading (index)
- **BNR**: Banca Națională a României (the National Bank of Romania)
- **BVB**: Bursa de Valori București (the Bucharest Stock Exchange)
- **CBB**: Circular Block Bootstrap
- **CEE**: Central and Eastern Europe
- **CLT**: Central Limit Theorem
- **EODHD**: EOD Historical Data (market-data provider)
- **EWC**: Equal-Weighted Cosine (estimator)
- **FWER**: Family-Wise Error Rate
- **GARCH**: Generalised AutoRegressive Conditional Heteroskedasticity
- **GDP**: Gross Domestic Product
- **HAC**: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- **HAR**: Heterogeneous AutoRegressive (model)
- **LLSW**: Lazarus, Lewis, Stock and Watson (2018)
- **MA**: Moving Average
- **MBB**: Moving-Block Bootstrap
- **MCMMP**: Metoda celor mai mici pătrate (ordinary least squares (OLS))
- **NW**: Newey–West (standard errors)
- **OLS**: Ordinary Least Squares
- **QS**: Quadratic Spectral (kernel)
- **SE**: Standard Error
- **SPA**: Superior Predictive Ability (test)
- **TSA**: Time Series Analysis

Today's question and seminar structure

- The question of the seminar
  - **when data are dependent, how wrong is the textbook standard error, and how do we fix it?**
- Outputs to produce
  - the answers to the A exercises
  - one figure and one table for each B exercise
  - the analysis requested in C1 and the verdicts on the AI answer in C2, each with one line of justification
  - one sentence of interpretation for each exercise
- Notebook for today: open the seminar notebook in Google Colab
  - each task names its notebook section
- Exercises, their types and their order: the exercise map (next slide)
- Seminar: Prof. Daniel Traian Pele, danpele@ase.ro

Exercise map

Task	Question	Type	Model
A1	long-run variance of AR(1) and MA(1); size of the naive test	Solved	–
A2, A3, A4	Newey–West by hand; a negative truncated estimate; why Bartlett is nonnegative	Solved, Proposed	A2
A5, A6	moving-block and circular bootstrap by hand	Solved, Proposed	A5
A7, A8	family-wise error, Bonferroni and Holm	Solved, Proposed	A7
B1, B2	HAC inference for a mean: S&P 500, Romanian GDP	Solved, Proposed	B1
B3, B4	Monte Carlo of size: AR(1) and GARCH(1,1)	Solved, Proposed	B3
B5, B6	replication of Granger and Newbold (1974); block bootstrap of EUR/RON	Solved, Proposed	B5, B1
C1, C2	data snooping on the BET; what is wrong in an AI answer?	Proposed	A7, B1–B6

- **[Solved]**: full solution in the slides and in the notebook, a model to follow; **[Proposed]**: you solve it, following the model; its solution is discussed in the seminar
- **A**: calculations and derivations on paper; **B**: real data, solved in the notebook, each closing with an interpretation question; **C**: open questions and the critique of an AI answer

Learning outcomes

- Compute a long-run variance from a model and from data, by hand and in code
- Explain why the Bartlett kernel gives a nonnegative variance and the truncated kernel may not
- Run a block bootstrap and choose its block length
- Measure the actual size of a test by Monte Carlo and report its Monte Carlo error
- Replicate a published simulation result and explain the gap

Data used

Series	Source	Frequency and unit	Period
S&P 500, BET	EODHD	daily log returns, %	2000–2026
real GDP, Romania	Eurostat (namq_10_gdp)	quarterly, q/q growth, %	2000Q1–2026Q2
EUR/RON	BNR reference rate	daily log change, %	2005–2026
simulated AR(1), GARCH(1,1), random walks	–	–	seed 2026

- In the notebook: `acov_all`, `lrv`, `andrews_bw`, `ewc_lrv`, `fixed_b_table`, `politis_white`, `mbb_means`, `sb_indices`; no account or key is needed

Prerequisites for Today (1/24): dependent data

- Details: Chapter 0
- **Weak stationarity** of x_t, $t = 1, \ldots, T$: constant mean and autocovariances that depend only on the distance j

$$\mathbb{E}x_t = \mu, \qquad \gamma_j = \mathrm{Cov}(x_t, x_{t-j}), \qquad \rho_j = \gamma_j/\gamma_0$$

  - γ_0: the variance; γ_j: the **autocovariance** at lag j; $\rho_j \in [-1, 1]$: the **autocorrelation** (ACF)
  - $\rho_j = 0$ for all $j \geq 1$: no linear dependence between dates
- Today's quantity: the **sample mean** $\bar{x} = T^{-1} \sum_{t=1}^{T} x_t$ and its precision
  - annual rate of a mean of log changes: the mean per period $\times A$, with A the number of periods per year (252 trading days, 4 quarters)

Prerequisites for Today (2/24): AR(1) and MA(1)

- **AR(1)**: today is a fraction ϕ of yesterday plus a new shock

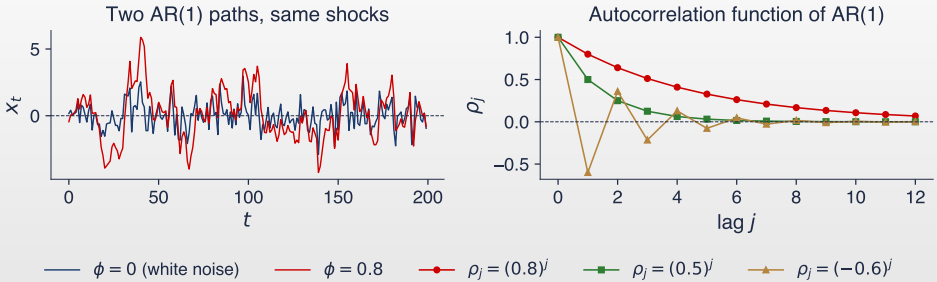
$$x_t = \phi x_{t-1} + \varepsilon_t, \qquad |\phi| < 1$$

  - ε_t: white noise (mean 0, variance σ^2, uncorrelated); ϕ: the persistence
  - $\gamma_0 = \sigma^2/(1-\phi^2)$ and $\rho_j = \phi^{|j|}$: the ACF decays geometrically
- **MA(1)**: today's shock plus a fraction θ of yesterday's shock

$$y_t = \varepsilon_t + \theta \varepsilon_{t-1}$$

  - $\gamma_0 = (1+\theta^2)\sigma^2$, $\gamma_1 = \theta\sigma^2$, $\gamma_j = 0$ for $|j| \geq 2$: a memory of one period
- **ARMA**(p, q) (TSA, Chapter 2): $\phi(L)x_t = \theta(L)\varepsilon_t$
  - L: the lag operator, $Lx_t = x_{t-1}$
  - $\phi(L) = 1 - \phi_1 L - \cdots - \phi_p L^p$, $\theta(L) = 1 + \theta_1 L + \cdots + \theta_q L^q$

Prerequisites for Today (3/24): simulated AR(1) paths and their ACF



- Left: the same shocks with $\phi = 0$ and $\phi = 0.8$; with $\phi = 0.8$ the path makes long excursions away from 0
- Right: the theoretical ACF $\rho_j = \phi^j$
  - $\phi > 0$: neighbouring observations repeat much of the same information; $\phi < 0$: the sign alternates

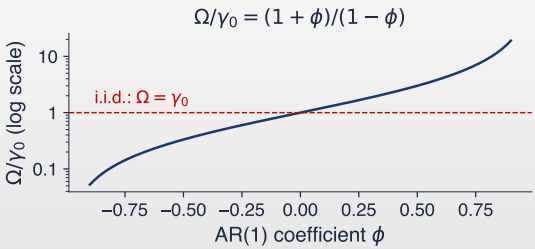
Prerequisites for Today (4/24): the variance of the mean

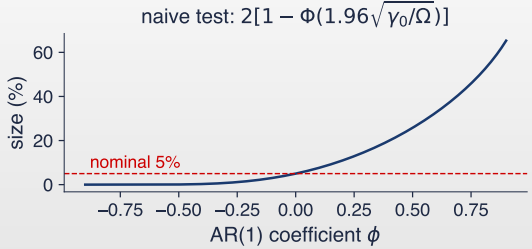
- The variance of $\bar{x}$ adds the covariances of **all pairs** of observations

$$\mathrm{Var}(\sqrt{T}\,\bar{x}) = \sum_{|j|<T} \Big(1 - \frac{|j|}{T}\Big)\gamma_j \;\longrightarrow\; \Omega = \gamma_0 + 2\sum_{j\geq 1}\gamma_j$$

  - $T - |j|$ pairs lie at distance j, hence the weight $1 - |j|/T$; as T grows, every weight tends to 1
- Ω: the **long-run variance**
  - $\Omega = 2\pi f(0)$, with $f(\omega) = (2\pi)^{-1}\sum_j \gamma_j e^{-ij\omega}$ the spectral density: only the frequency zero matters for a mean
  - so $\mathrm{Var}(\bar{x}) \approx \Omega/T$, not γ_0/T
- Reading the ratio Ω/γ_0
  - $= 1$: i.i.d. data or a martingale difference; the textbook standard error $\sqrt{\hat{\gamma}_0/T}$ is right
  - > 1 (positive dependence): the naive standard error is too small; < 1 (negative dependence): too large

Prerequisites for Today (5/24): the long-run variance of an ARMA





- ARMA: replace L by 1 in the two polynomials

$$\Omega = \sigma^2 \frac{\theta(1)^2}{\phi(1)^2}, \qquad \phi(1) = 1 - \textstyle\sum_i \phi_i, \quad \theta(1) = 1 + \textstyle\sum_i \theta_i$$

  - AR(1): $\Omega = \sigma^2/(1-\phi)^2$ and $\Omega/\gamma_0 = (1+\phi)/(1-\phi)$ (left)
  - right: the rejection rate of the naive 5% test of a true mean, derived on the next slide

Prerequisites for Today (6/24): CLT, t-test and size

- **CLT** (central limit theorem) for dependent data

$$\sqrt{T}(\bar{x} - \mu) \xrightarrow{d} N(0, \Omega)$$

  - $\xrightarrow{d}$: convergence in distribution; holds when dependence dies out fast enough (mixing) and moments exist, or for martingale differences ($\Omega = \gamma_0$)

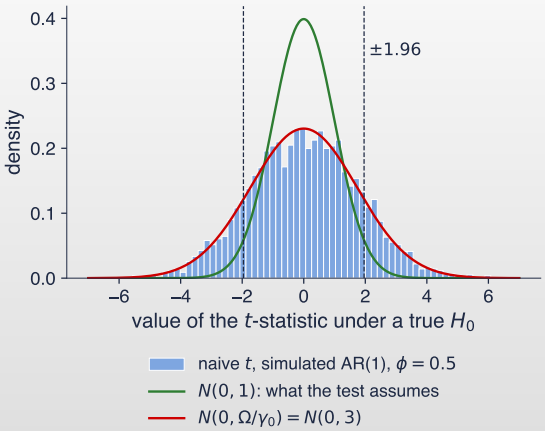
- t-**statistic** for $H_0 : \mu = \mu_0$

$$t = \frac{\bar{x} - \mu_0}{\sqrt{\hat{\Omega}/T}}$$

  - reject at 5% if $|t| > 1.96$; p-value $2[1 - \Phi(|t|)]$, with Φ the distribution function of $N(0,1)$
  - the **naive** test uses $\hat{\Omega} = \hat{\gamma}_0$; then $t \to_d N(0, \Omega/\gamma_0)$ and $P(|t| > 1.96) \to 2[1 - \Phi(1.96\sqrt{\gamma_0/\Omega})]$

- **Size**: the rejection rate when H_0 is true; **power**: the rejection rate when H_0 is false
  - size above the nominal 5%: the test is **oversized**; below: **conservative**

Prerequisites for Today (7/24): the naive t under dependence



- 4000 simulated AR(1) samples, $\phi = 0.5$, $T = 200$, true mean 0
- The naive t spreads like $N(0, 3)$
  - $\Omega/\gamma_0 = 1.5/0.5 = 3$
  - the test assumes the green curve, $N(0, 1)$
- Share beyond ± 1.96
  - $2[1 - \Phi(1.96/\sqrt{3})] \approx 26\%$ instead of 5%
  - a true H_0 is rejected five times too often

Prerequisites for Today (8/24): kernel (HAC) estimators

- **Sample autocovariance** (divisor T)

$$\hat{\gamma}_j = \frac{1}{T} \sum_{t=j+\mathbf{1}}^{T} (x_t - \bar{x})(x_{t-j} - \bar{x})$$

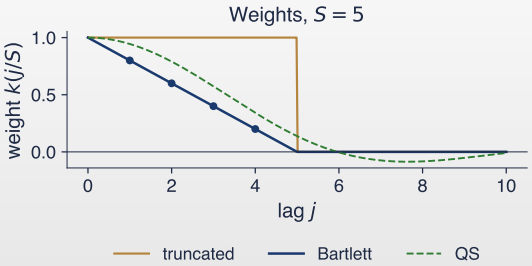
  - dividing by T, not $T - j$, keeps the estimated autocovariances a valid (nonnegative definite) sequence

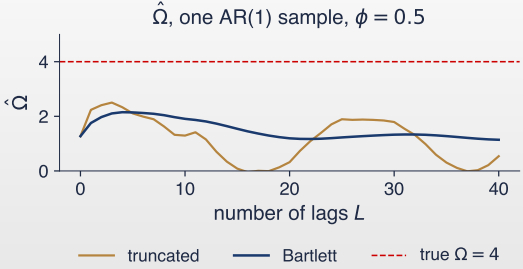
- **Kernel estimator** of Ω: a weighted sum of sample autocovariances

$$\hat{\Omega} = \hat{\gamma}_{\mathbf{0}} + 2 \sum_{j=\mathbf{1}}^{T-\mathbf{1}} k(j/S)\, \hat{\gamma}_j$$

  - $k(\cdot)$: the kernel (weight function, $k(0) = 1$, decreasing towards 0); S: the **bandwidth**
  - HAC (heteroskedasticity and autocorrelation consistent): $\hat{\Omega} \to \Omega$ if $S \to \infty$ and $S/T \to 0$
  - small S: bias (distant covariances ignored); large S: variance (many noisy $\hat{\gamma}_j$)

Prerequisites for Today (9/24): three kernels





- **Bartlett** (Newey–West, Newey and West (1987)): $k(x) = 1 - |x|$ for $|x| \leq 1$
  - with $S = L + 1$: weights $1 - j/(L+1)$ for the lags $j = 1, \ldots, L$
- **Truncated**: $k(x) = 1$ for $|x| \leq 1$, i.e. $\hat{\Omega} = \hat{\gamma}_0 + 2\sum_{j \leq L} \hat{\gamma}_j$ (right: it jumps with L)
- **Quadratic spectral** (QS): smooth, gives a weight to every lag

$$k(x) = \frac{25}{12\pi^2 x^2}\left[\frac{\sin(6\pi x/5)}{6\pi x/5} - \cos(6\pi x/5)\right]$$

Prerequisites for Today (10/24): choosing the bandwidth

- **Rule of thumb** Newey and West (1994)

$$L = \lfloor 4(T/100)^{2/9} \rfloor, \qquad S = L + 1$$

  - $\lfloor \cdot \rfloor$: the integer part; for example $T = 100$ gives $L = 4$

- **Andrews** Andrews (1991): the S that minimises the mean squared error of $\hat{\Omega}$ for an AR(1)

$$S = 1.1447\,(\hat{\alpha}\,T)^{1/3}, \qquad \hat{\alpha} = \frac{4\hat{\rho}^2}{(1-\hat{\rho})^2(1+\hat{\rho})^2}$$

  - $\hat{\rho}$: the estimated AR(1) coefficient of the series; more persistence gives a larger S
  - $\hat{\rho} \approx 0$: S close to 0, so almost no lag gets a weight: the naive variance

- **LLSW** Lazarus et al. (2018): Newey–West with $S = 1.3\sqrt{T}$, a much larger bandwidth
  - less bias, more noise in $\hat{\Omega}$; the noise is paid for with **fixed-b** critical values Kiefer and Vogelsang (2005)

Prerequisites for Today (11/24): the EWC estimator

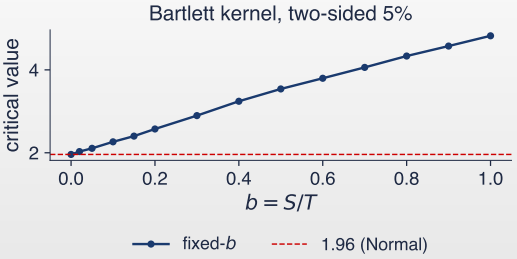
- **EWC** (equal-weighted cosine): project the demeaned series on ν slow cosines

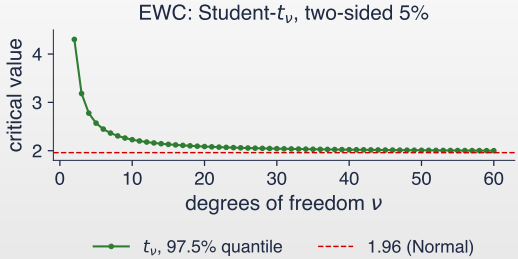
$$\hat{\Lambda}_j = \sqrt{\tfrac{2}{T}} \sum_{t=1}^{T} \cos\left[\pi j \, \frac{t-1/2}{T}\right](x_t - \bar{x}), \qquad \hat{\Omega}_{\mathrm{EWC}} = \frac{1}{\nu} \sum_{j=1}^{\nu} \hat{\Lambda}_j^2$$

  - $\hat{\Lambda}_j$: the weight of the j-th cosine (period $2T/j$) in the data; the low frequencies carry Ω: $\mathbb{E}\hat{\Lambda}_j^2 \approx \Omega$ for small j
  - $\nu = 0.4\, T^{2/3}$ Lazarus et al. (2018); for example $T = 100$ gives $\nu \approx 9$

- $\hat{\Omega}_{\mathrm{EWC}}$ behaves like $\Omega\, \chi_\nu^2/\nu$, so $t \approx t_\nu$
  - χ_ν^2: a sum of ν squared independent $N(0,1)$; $t_\nu = N(0,1)/\sqrt{\chi_\nu^2/\nu}$: the Student distribution with ν degrees of freedom
  - critical values from t_ν, larger than 1.96: the price of a noisy variance estimate

Prerequisites for Today (12/24): fixed-b and Student critical values



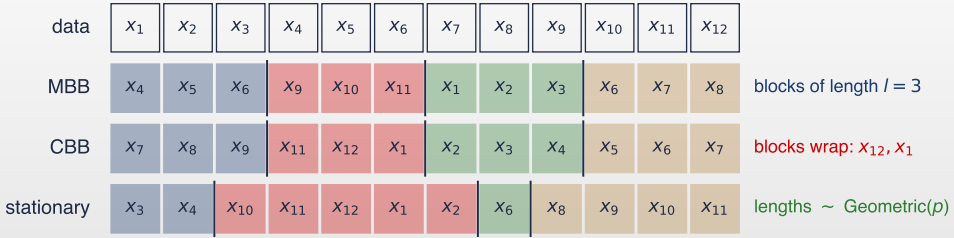


- **Fixed-b** Kiefer and Vogelsang (2005): keep $b = S/T$ fixed as $T \to \infty$
  - $\hat{\Omega}$ stays random, so t has a non-Normal limit; its 5% critical value grows with b (left, simulated)
- EWC: the critical value of t_ν falls towards 1.96 as ν grows (right)

Prerequisites for Today (13/24): block bootstraps

- **Bootstrap**: build an artificial series $x_1^*, \ldots, x_T^*$ from the data and repeat B times
  - a star marks a bootstrap quantity: $\bar{x}^*$, and $\mathbb{E}^*$, Var^* (mean and variance over resamples, given the data)
  - the i.i.d. bootstrap draws single observations: it destroys the dependence, $\mathrm{Var}^*(\bar{x}^*) \approx \hat{\gamma}_0 / T$
- **Moving blocks** (MBB) Künsch (1989): blocks $B_i = (x_i, \ldots, x_{i+l-1})$, $i = 1, \ldots, T - l + 1$
  - draw $k = \lceil T/l \rceil$ blocks with replacement and concatenate them ($\lceil \cdot \rceil$: rounding up)
  - the dependence inside each block is kept; consistent if $l \to \infty$ and $l/T \to 0$
- **Circular** (CBB): set $x_{T+i} = x_i$, so that blocks may wrap around the end; T blocks
- **Stationary** Politis and Romano (1994): circular blocks with random lengths
  - lengths i.i.d. Geometric(p): $P(\text{length} = m) = (1-p)^{m-1} p$, mean $1/p$

Prerequisites for Today (14/24): three block-bootstrap schemes



- $T = 12$; each colour is one block drawn from the data; vertical bars: joins between blocks
- MBB: $x_{\mathbf{1}}$ and $x_{\mathbf{12}}$ belong to only one block each, the middle values to three: an **edge effect**
- Stationary bootstrap: here four blocks of random lengths 2, 5, 1 and 4

Prerequisites for Today (15/24): bootstrap intervals and block length

- From B resample means $\bar{x}_1^*, \ldots, \bar{x}_B^*$, with q_a^* their a-quantile
  - bootstrap standard error: the standard deviation of the $\bar{x}_b^*$
  - **percentile** 95% interval $[q^*_{0.025},\ q^*_{0.975}]$; **basic** interval $[2\bar{x} - q^*_{0.975},\ 2\bar{x} - q^*_{0.025}]$
- **Politis–White** block length Politis and White (2004), Patton, Politis and White (2009)

$$\hat{l}_{\mathrm{SB}} = \Big(\frac{2\hat{G}^2}{\hat{D}}\Big)^{1/3} T^{1/3}, \quad \hat{G} = \sum_{|k|\le M} \lambda\Big(\frac{k}{M}\Big)|k|\hat{\gamma}_k, \quad \hat{D} = 2\hat{g}^2, \quad \hat{g} = \sum_{|k|\le M} \lambda\Big(\frac{k}{M}\Big)\hat{\gamma}_k$$

  - λ: a flat-top weight (1 up to $|x| = 1/2$, then linear to 0 at 1); M: twice the lag after which the sample ACF stays insignificant
  - CBB: $\hat{D} = \frac{4}{3}\hat{g}^2$; more autocorrelation (larger $|\hat{G}|$) gives longer blocks; almost no autocorrelation gives $\hat{l} \approx 1$, the i.i.d. bootstrap

Prerequisites for Today (16/24): wild bootstrap and Monte Carlo

- **Wild bootstrap**: keep every observation in place and flip its deviation at random

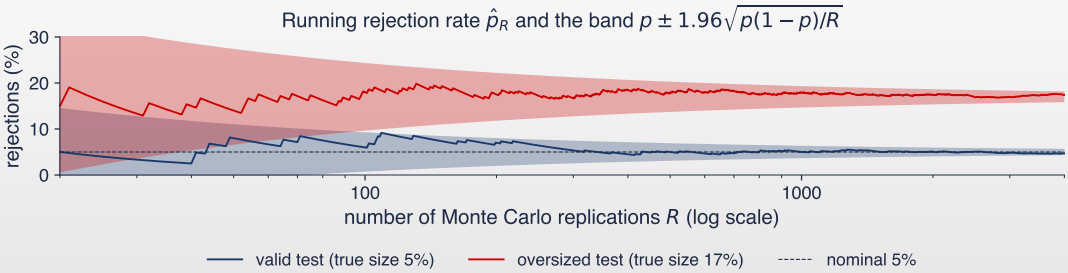
$$x_t^* = \bar{x} + \eta_t (x_t - \bar{x}), \qquad \eta_t = \pm 1 \text{ with probability } 1/2$$

  - keeps the size of each deviation (heteroskedasticity); destroys the autocorrelation, since the η_t are independent
- **Monte Carlo** study of size: simulate R samples under H_0, run the test on each

$$\hat{p} = \frac{1}{R} \sum_{r=1}^{R} 1\{|t_r| > c\}, \qquad \mathrm{se}(\hat{p}) = \sqrt{\hat{p}(1-\hat{p})/R}$$

  - $1\{\cdot\}$: 1 if the condition holds, 0 otherwise; t_r: the statistic in sample r; c: the critical value
  - report R, the seed and $\mathrm{se}(\hat{p})$; drop a burn-in (the first simulated values) so that the start value does not matter

Prerequisites for Today (17/24): the Monte Carlo error



- Each line: the share of rejections after the first R replications; band: $p \pm 1.96\,\mathrm{se}$
- With $R = 100$ a true 5% size can show up anywhere between about 1% and 9%; with $R = 4000$ the band is ± 0.7 pp

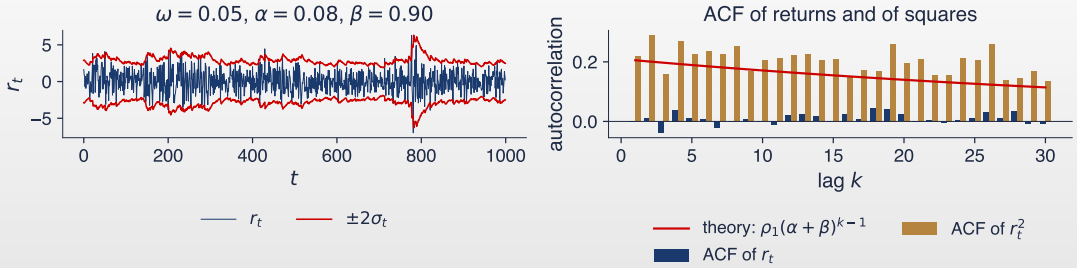
Prerequisites for Today (18/24): the GARCH(1,1) model

- **GARCH(1,1)**: the variance of today's return depends on yesterday's squared return and variance

$$r_t = \sigma_t z_t, \qquad \sigma_t^2 = \omega + \alpha r_{t-1}^2 + \beta \sigma_{t-1}^2$$

  - z_t: i.i.d. innovations, mean 0, variance 1; σ_t^2: the conditional variance; $\omega > 0$, $\alpha, \beta \geq 0$
  - α: reaction to news; β: memory of past variance; $\alpha + \beta$: the persistence
- Moments
  - $\alpha + \beta < 1$: stationary, with unconditional variance $\text{Var} r_t = \omega/(1 - \alpha - \beta)$
  - with Normal z_t, $\mathbb{E} r_t^4 < \infty$ if and only if $3\alpha^2 + 2\alpha\beta + \beta^2 < 1$
- r_t is a **martingale difference**: $\mathbb{E}(r_t \mid \text{past}) = 0$
  - so r_t is uncorrelated; but r_t^2 is autocorrelated, $\rho_k(r_t^2) = \rho_1(\alpha + \beta)^{k-1}$, with ρ_1 the first autocorrelation of r_t^2

Prerequisites for Today (19/24): volatility clustering in a simulated GARCH



- Left: calm and turbulent periods alternate; the band $\pm 2\sigma_t$ follows them
- Right: the returns are uncorrelated, the squares are not; their ACF decays at the rate $\alpha + \beta = 0.98$

Prerequisites for Today (20/24): random walks and spurious regression

- **Random walk**: the accumulated sum of i.i.d. shocks

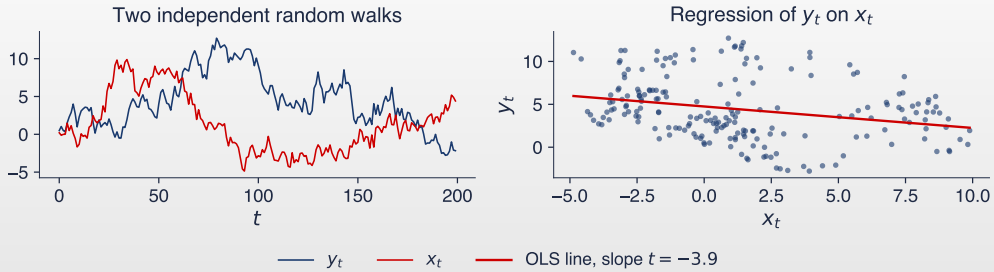
$$y_t = y_{t-1} + e_t, \qquad \mathrm{Var}(y_t - y_0) = t\sigma_e^2$$

  - a unit root ($\phi = 1$): not stationary, the variance grows with t; $\Delta y_t = y_t - y_{t-1} = e_t$ is i.i.d.
- OLS regression $y_t = a + bx_t + u_t$ and the t-statistic of the slope

$$t_{\hat{b}} = \frac{\hat{b}}{\mathrm{se}(\hat{b})}, \qquad \mathrm{se}(\hat{b}) = \sqrt{\frac{\hat{\sigma}_u^2}{\sum_t (x_t - \bar{x})^2}}$$

  - OLS (MCMMP): ordinary least squares; $\hat{\sigma}_u^2$: the residual variance
  - **spurious regression** Granger and Newbold (1974): two independent random walks often give $|t_{\hat{b}}| > 2$
  - the residuals u_t are themselves a random walk, so $t_{\hat{b}}$ grows like $\sqrt{T}$ instead of settling on $N(0,1)$ Phillips (1986)

Prerequisites for Today (21/24): two independent random walks



- Left: y_t and x_t are simulated with independent shocks; each wanders far from its start
- Right: the OLS line looks highly significant although there is no relation at all

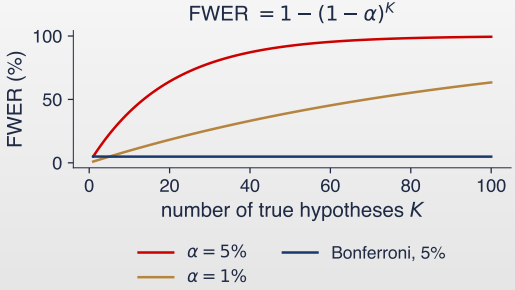
Prerequisites for Today (22/24): multiple testing

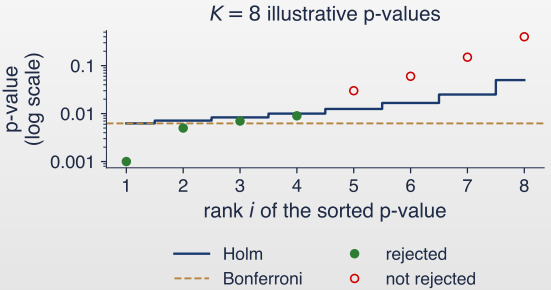
- K tests, each at level α; **FWER** (family-wise error rate): the probability of at least one false rejection

$$\mathrm{FWER} = 1 - (1-\alpha)^K \quad \text{(independent tests, all } H_0 \text{ true)}$$

  - the expected number of false rejections is $K\alpha$
- **Bonferroni**: test each hypothesis at α/K
  - FWER $\leq K \cdot \alpha/K = \alpha$ under any dependence (the probability of a union is at most the sum)
- **Holm** Holm (1979): sort the p-values $p_{(1)} \leq \cdots \leq p_{(K)}$
  - reject $H_{(i)}$ while $p_{(i)} \leq \alpha/(K-i+1)$; stop at the first failure
  - adjusted p-values $\tilde{p}_{(i)} = \max_{j \leq i} \min\{1, (K-j+1)p_{(j)}\}$: reject when $\tilde{p}_{(i)} \leq \alpha$

Prerequisites for Today (23/24): FWER, Bonferroni and Holm





- Left: at 5% each, 14 independent true hypotheses already give a chance above 50% of a false discovery
- Right: the Holm thresholds start at Bonferroni and relax step by step; here Holm rejects four hypotheses, Bonferroni two

Prerequisites for Today (24/24): data snooping and the Reality Check

- Trading rules on prices P_t: position $s_t \in \{0, 1\}$ (1 = long), rule return $s_t r_{t+1}$
  - moving average: $s_t = 1\{P_t > \mathrm{MA}_n(t)\}$, $\mathrm{MA}_n(t) = n^{-1} \sum_{i=0}^{n-1} P_{t-i}$; momentum: $s_t = 1\{P_t > P_{t-n}\}$
  - $f_{k,t} = s_{k,t} r_{t+1} - r_{t+1}$: the excess return of rule k over buy-and-hold; $\bar{f}_k$: its mean over n days
- **Data snooping**: the best of K rules looks significant by chance; its naive t ignores the search
- **Reality Check** White (2000): H_0: no rule beats the benchmark, $\max_k \mathbb{E} f_{k,t} \leq 0$

$$V = \max_k \sqrt{n}\, \bar{f}_k, \qquad V_b^* = \max_k \sqrt{n}\, (\bar{f}_{k,b}^* - \bar{f}_k)$$

  - $\bar{f}_{k,b}^*$: from the stationary bootstrap of the days, the same days for all rules; p-value: the share of $V_b^* > V$

A1: long-run variances and the naive test [Solved]

Task

- AR(1) $x_t = 0.6x_{t-1} + \varepsilon_t$ and MA(1) $y_t = \varepsilon_t + 0.5\varepsilon_{t-1}$, $\sigma^2 = 1$.
- 1. Compute γ_0, Ω and Ω/γ_0 for both processes.
- 2. Compute the asymptotic size of the naive 5% t-test for the AR(1) mean.
- 3. Compute Ω/γ_0 for $\phi = -0.6$ and say what happens to the naive test.
- Report: two ratios, one size, one sentence.

Solution

- 1. AR(1): $\gamma_0 = 1/(1 - 0.36) = 1.5625$, $\Omega = 1/0.4^2 = 6.25$, ratio $1.6/0.4 = 4.00$
- MA(1): $\gamma_0 = 1 + \theta^2 = 1.25$, $\gamma_1 = \theta = 0.50$, $\Omega = (1 + \theta)^2 = 2.25$, ratio 1.80
- 2. The naive t is $N(0, \Omega/\gamma_0) = N(0, 4)$: size $= 2[1 - \Phi(1.96/2)] = 32.7\%$
- 3. $\phi = -0.6$: ratio $0.4/1.6 = 0.25$; the naive test almost never rejects: it is conservative and has little power

Back Exercise list

A2: Newey–West by hand [Solved]

Task

- An illustrative series of $T = 10$ values: $(2.1, 1.4, 0.3, -0.5, 0.2, 1.1, 1.8, 0.9, -0.2, 0.4)$.
- 1. Compute $\bar{x}$, $\hat{\gamma}_0$, $\hat{\gamma}_1$, $\hat{\gamma}_2$ with divisor T.
- 2. Compute the Newey–West $\hat{\Omega}$ with $L = 2$ (the rule of thumb for $T = 10$).
- 3. Compare the naive and the HAC t-statistics of $H_0 : \mu = 0$.
- Report: three autocovariances, $\hat{\Omega}$, two t-statistics.

Solution

- 1. $\bar{x} = 0.7500$; $\hat{\gamma}_0 = 0.6585$, $\hat{\gamma}_1 = 0.2357$, $\hat{\gamma}_2 = -0.3185$
- 2. Weights $1 - 1/3 = 2/3$ and $1 - 2/3 = 1/3$: $\hat{\Omega} = \hat{\gamma}_0 + 2(\frac{2}{3}\hat{\gamma}_1 + \frac{1}{3}\hat{\gamma}_2) = 0.7605$
- 3. SE naive $\sqrt{\hat{\gamma}_0/10} = 0.257$, $t = 2.923$; SE HAC $\sqrt{\hat{\Omega}/10} = 0.276$, $t = 2.720$
- The positive $\hat{\gamma}_1$ and the negative $\hat{\gamma}_2$ partly cancel; with $T = 10$ the estimate is very noisy (fixed-b would raise the critical value)

Back Exercise list

A3: a negative variance [Proposed]

Solution discussed in class

Task

- The series $(1, -1, 1, -1, 1, -1, 1, -1, 2, -2)$. Model: A2.
- 1. Compute $\hat{\gamma}_0$, $\hat{\gamma}_1$, $\hat{\gamma}_2$.
- 2. Compute the truncated-kernel estimate with $L = 1$ and the Newey–West estimate with $L = 1$.
- 3. Compute the Newey–West estimate with $L = 2$.
- Report: three autocovariances, three variance estimates, one sentence on the sign.

A4: why the Bartlett kernel is nonnegative [Proposed]

Solution discussed in class

Task

- Demeaned data $u_1, \dots, u_T$; Newey–West with L lags. Model: A2.
- 1. Let $s_t = \sum_{i=0}^{L} u_{t-i}$ (with $u_t = 0$ outside the sample). Show that $\frac{1}{(L+1)T} \sum_{t=1}^{T+L} s_t^2 = \hat{\gamma}_0 + 2 \sum_{j=1}^{L} (1 - \frac{j}{L+1}) \hat{\gamma}_j$.
- 2. Conclude that $\hat{\Omega}_{NW} \geq 0$ for every sample.
- 3. Explain why no such representation exists for the truncated kernel.
- Report: the derivation and one sentence.

Back Exercise list

A5: the moving-block bootstrap by hand [Solved]

Task

- Series $x = (1, 3, 2, 6, 4, 2)$, block length $l = 2$, so $k = 3$ blocks per resample.
- 1. List the moving blocks and their means.
- 2. Compute $E^*\bar{x}^*$ and compare it with $\bar{x}$.
- 3. Compute $\mathrm{Var}^*(\bar{x}^*)$ and compare it with $\hat{\gamma}_0/T$ and with the Bartlett estimate $\hat{\Omega}(S = 2)/T$.
- Report: five block means, two expectations, three variances.

Solution

- 1. Blocks $(1, 3), (3, 2), (2, 6), (6, 4), (4, 2)$; means $2.0, 2.5, 4.0, 5.0, 3.0$
- 2. $\bar{x}^*$ is the average of 3 block means drawn uniformly: $E^*\bar{x}^* = 3.3000 \neq \bar{x} = 3.0000$: the end values 1 and 2 enter only one block each (edge bias)
- 3. $\mathrm{Var}^*(\bar{x}^*) = \mathrm{Var}^*(\text{block mean})/3 = 0.3867$; naive $\hat{\gamma}_0/6 = 0.4444$; Bartlett 0.4167
- With $T = 6$ the three numbers differ; asymptotically MBB with block l and Bartlett with $S = l$ coincide Künsch (1989)

Back Exercise list

A6: the circular block bootstrap by hand [Proposed]

Solution discussed in class

Task

- The same series and $l = 2$; wrap the series on a circle. Model: A5.
- 1. List the six circular blocks and their means.
- 2. Show that $E^*\bar{x}^* = \bar{x}$ and explain why.
- 3. Compute $\mathrm{Var}^*(\bar{x}^*)$.
- Report: six block means, one expectation, one variance, one sentence.

Back Exercise list

A7: the family-wise error [Solved]

Task

- A researcher tests 20 independent, useless predictors of inflation, each at 5%.
- 1. Compute the probability of at least one rejection (FWER).
- 2. Compute the expected number of false rejections.
- 3. Give the Bonferroni level and the resulting FWER.
- Report: two probabilities, one count, one level.

Solution

- 1. FWER $= 1 - 0.95^{\mathbf{20}} = 64.2\%$
- 2. $E[\#\text{false rejections}] = 20 \cdot 0.05 = 1$
- 3. Bonferroni: $0.05/20 = 0.0025$; FWER $= 1 - (1 - 0.0025)^{\mathbf{20}} = 4.88\% \leq 5\%$
- Bonferroni holds under any dependence (union bound); with positively correlated tests it is conservative

Back Exercise list

A8: Holm step-down [Proposed]

Solution discussed in class

Task

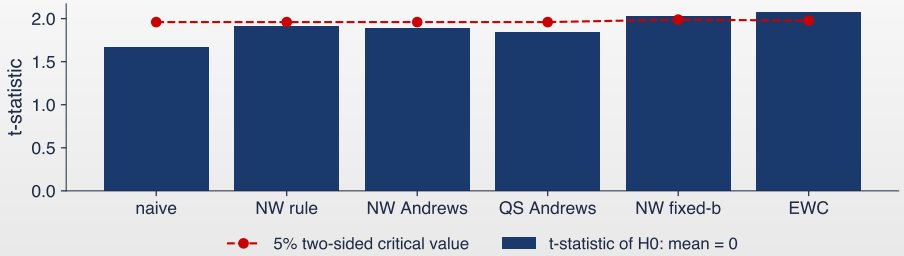
- Six sorted p-values: $0.001, 0.008, 0.012, 0.030, 0.041, 0.240$; FWER target 5%. Model: A7.
- 1. Write the six Holm thresholds $0.05/(6 - i + 1)$.
- 2. Count the rejections with Holm, with Bonferroni and with no correction.
- 3. With 50 tests at 1%, give the FWER of uncorrected tests and the Bonferroni level.
- Report: six thresholds, three counts, two numbers.

Back Exercise list

B1: the mean return of the S&P 500 [Solved]

- **Question**: is the mean daily log return of the S&P 500 since 2000 significantly positive, and does the answer depend on the standard error?
- **Inputs**: S&P 500 daily log returns in %, 2000–2026 ($T = 6\,714$)
- **Tasks**
  1. Compute the mean and $\hat{\rho}_1$.
  2. Compute five standard errors: naive, Newey–West rule of thumb, Newey–West with the Andrews bandwidth, NW with $S = 1.3\sqrt{T}$ (fixed-b), EWC.
  3. Compute each t-statistic for $H_0 : \mu = 0$ and compare it with its critical value.
  4. Interpretation: why are the robust standard errors smaller than the naive one here?
- **Report**: one mean, one autocorrelation, five standard errors with their t-statistics, one sentence
- Notebook: section B1

B1: solution [Solved]



- $\bar{r} = 0.0247\%$ a day (6.23% a year), $\hat{\rho}_1 = -0.099$; SE: naive 0.0148, NW rule ($S = 11$) 0.0129, NW Andrews ($S = 7.4$) 0.0131, fixed-b ($S = 106.5$) 0.0122, EWC ($\nu = 142$) 0.0119
- t: naive 1.67, NW 1.91, Andrews 1.88 (all against 1.96); fixed-b 2.03 against 1.99; EWC 2.07 against 1.98: only the last two reject, barely
- Interpretation: the negative first-order autocorrelation makes $\Omega < \gamma_0$; daily index returns mean-revert slightly, so the naive test is conservative; the conclusion stays fragile

ATS_ch0_seminar

B2: the mean growth of Romanian GDP [Proposed]

- **Question**: how precisely do we know the average quarterly growth of Romanian real GDP since 2000?
- **Inputs**: Eurostat real GDP, seasonally adjusted, q/q log growth in %, 2000Q1–2026Q2 ($T = 106$); model: B1
- **Tasks**
  1. Compute the mean, $\hat{\rho}_1$ and the 95% intervals with the five methods of B1.
  2. Add a stationary-bootstrap percentile interval with the Politis–White block length (4999 resamples).
  3. Repeat the main estimates without the four quarters of 2020.
  4. Interpretation: what matters more for this mean, dependence or the 2020 quarters?
- **Report**: one mean, six intervals, one block length, one comparison, one sentence
- Notebook: section B2

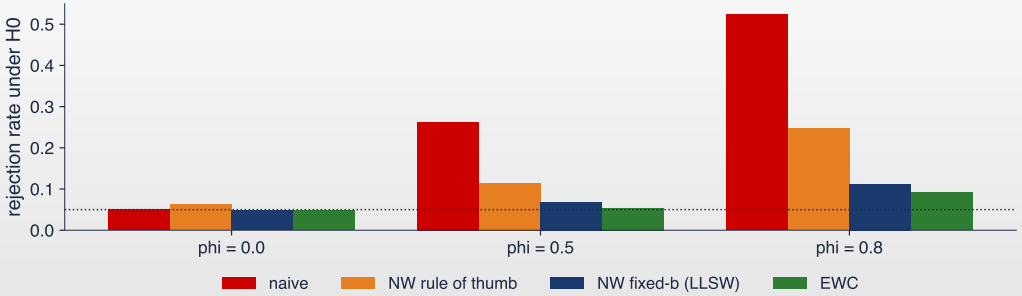
Back Exercise list

B3: Monte Carlo of size under AR(1) [Solved]

- **Question**: with a sample as long as the Romanian GDP series, which test of a mean keeps its 5% size when the data are autocorrelated?
- **Inputs**: simulated AR(1), $\phi \in \{0, 0.5, 0.8\}$, $T = 106$, 4 000 replications, seed 2026
- **Tasks**
  1. Simulate the AR(1) paths with 200 burn-in values and $N(0, 1)$ innovations.
  2. For each path test $H_0 : \mu = 0$ at 5% with the naive test, NW rule of thumb, NW fixed-b and EWC.
  3. Report the rejection rates and their Monte Carlo standard errors.
  4. Interpretation: which test would you use for the mean of a series with $\hat{\rho}_1 = 0.5$ and $T \approx 100$?
- **Report**: a 3×4 table of rejection rates, one Monte Carlo error, one sentence
- Notebook: section B3

Back Exercise list

B3: solution [Solved]



- $\phi = 0$: naive 5.0%, NW 6.2%, fixed-b 4.8%, EWC 4.7%; $\phi = 0.5$: 26.3%, 11.4%, 6.9%, 5.4%; $\phi = 0.8$: 52.3%, 24.8%, 11.2%, 9.1%
- Monte Carlo standard error: $\sqrt{0.05 \cdot 0.95/4000} \approx 0.34$ pp at 5%, about 0.8 pp at 50%
- Interpretation: at $\phi = 0.5$ and $T \approx 100$ use EWC or NW fixed-b; the naive test and NW with the rule of thumb reject far too often, and the gap grows with ϕ

ATS_ch0_seminar

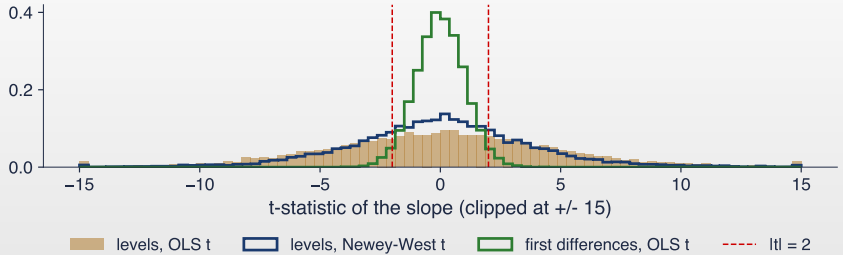
B4: Monte Carlo of size under GARCH(1,1) [Proposed]

- **Question**: does conditional heteroskedasticity alone distort a test on the mean of returns, and a test on the mean of squared returns?
- **Inputs**: simulated GARCH(1,1) of Prerequisites (18/24) with $N(0,1)$ innovations z_t, $\omega = 0.02$, $\alpha = 0.10$, $\beta = 0.88$, $T = 1\,000$, 4 000 replications; model: B3
- **Tasks**
  1. Check that the unconditional variance is 1 and that the fourth moment exists (the two conditions of Prerequisites (18/24)).
  2. Test $H_0 : Er_t = 0$ and $H_0 : Er_t^2 = 1$ at 5% with the four tests of B3.
  3. Compute ρ_1 of r_t^2 from the GARCH formula $\alpha(1 - \alpha\beta - \beta^2)/(1 - 2\alpha\beta - \beta^2)$.
  4. Interpretation: why is the first test fine and the second one not?
- **Report**: one moment condition, eight rejection rates, one autocorrelation, one sentence
- Notebook: section B4

B5: replicating Granger and Newbold (1974) [Solved]

- **Question**: how often does a regression of one random walk on an independent one look significant, and can HAC standard errors fix it?
- **Inputs**: Granger and Newbold (1974): $T = 50$, two independent Gaussian random walks, 100 regressions; published result: $|t| > 2$ in 76 of 100 cases
- **Tasks**
  1. Replicate the design: simulate y_t and x_t, regress y_t on $1, x_t$, record the OLS t of the slope; use first 100, then 10 000 replications.
  2. Recompute the share of $|t| > 2$ with Newey–West standard errors (rule of thumb) and with NW fixed-b.
  3. Repeat the regression in first differences.
  4. Interpretation: is the published 76% consistent with your replication?
- **Report**: four rejection shares, one Monte Carlo error, one sentence
- Notebook: section B5

B5: solution [Solved]



- First 100 replications: 68.0%; 10 000 replications: OLS 65.9%, Newey–West 54.2%, fixed-b 41.4% (critical value 2.48); first differences: 5.1%
- With 100 replications the Monte Carlo standard error is about 4.7 pp: 76% lies about two standard errors above our long-run rate, so the published number is a high draw, not a different phenomenon
- Interpretation: the result replicates in substance; HAC corrections reduce but do not remove the problem, because the t-statistic diverges Phillips (1986); differencing restores the 5% size

ATS_ch0_seminar

B6: block bootstrap of EUR/RON [Proposed]

- **Question**: has the leu depreciated significantly against the euro since 2005, on a daily mean basis?
- **Inputs**: EUR/RON BNR reference rate, daily log changes in %, 5 July 2005 to 18 September 2026 ($T = 5\,352$); model: B1
- **Tasks**
  1. Compute the mean, $\hat{\rho}_1$ and the Politis–White block lengths for the stationary and circular bootstrap.
  2. Compute bootstrap standard errors and 95% basic intervals with the i.i.d., moving-block, circular and stationary bootstrap (1999 resamples).
  3. Convert the mean into an average annual depreciation and compare it with the change of the rate between the first and the last day.
  4. Interpretation: why does the block bootstrap change so little here?
- **Report**: one mean, two block lengths, four intervals, one annual rate, one sentence
- Notebook: section B6

Back Exercise list

C1: data snooping on the BET [Proposed]

- **Question**: after adding momentum rules to the 50 moving-average rules of the lecture, does any rule beat buy-and-hold on the BET in 2014–2026?
- **Inputs**: BET daily closes (EODHD); rules: long if the close is above its n-day moving average ($n = 5, \ldots, 250$) or if the n-day return is positive ($n = 5, 10, 20, 60, 120, 250$); model: A7, B1
- **Tasks**
  1. Write down, before computing, the rule universe, the sample and the benchmark (a pre-analysis plan).
  2. Compute the excess return of each rule over buy-and-hold and the naive t of the best rule.
  3. Run the Reality Check White (2000) with the stationary bootstrap (mean block 10, 999 resamples).
  4. Interpretation: how would transaction costs of 0.1% per trade change the conclusion?
- **Report**: a dated plan, one t-statistic, one Reality Check p-value, one sentence; possible project seed
- Notebook: section C1

C2: critique of an AI answer [Proposed]

- A student asked an AI assistant: "How do I get correct standard errors for the mean of daily squared returns?" The answer contained six statements:
  - (a) "Newey–West standard errors are always larger than OLS standard errors."
  - (b) "For maximum robustness use $L = T - 1$ lags."
  - (c) "The wild bootstrap handles the autocorrelation of squared returns."
  - (d) "With $T = 6700$ observations the normal critical value 1.96 is always accurate."
  - (e) "The truncated kernel is better because it does not down-weight any lag."
  - (f) "The stationary bootstrap of Politis and Romano (1994) produces a stationary resampled series."
- Tasks
  - 1. For each statement, say whether it is correct; if not, give the correct statement and the exercise of today that shows it.
  - 2. Report: a list of six verdicts with one line of justification each.

Key takeaways

- The precision of a mean is set by $\Omega = \gamma_0 + 2\sum_j \gamma_j$; positive dependence inflates it, negative dependence deflates it
- Bartlett (Newey–West) is a sum of squares and never negative; the truncated kernel can be
- Block bootstraps resample dependence; the circular version removes the edge bias
- A Monte Carlo of size is the honest check of any test: report R, the seed and the Monte Carlo error
- A replication is successful when the gap is explained, not when the digits match

After the seminar

- Lecture 0 develops each topic of today: ergodicity and mixing, the HAC estimators, fixed-b, the block bootstraps, data snooping and reproducible research
- C1 can grow into a team project: data snooping in CEE technical-trading or macro-forecasting studies
- Reading: Newey and West (1987); Lazarus et al. (2018); Künsch (1989); Granger and Newbold (1974)
- **The seminar is for practice and is not graded**

References

- Andrews, D.W.K. (1991). Heteroskedasticity and autocorrelation consistent covariance matrix estimation. *Econometrica*, 59(3), 817–858.
- Andrews, D.W.K., Monahan, J.C. (1992). An improved heteroskedasticity and autocorrelation consistent covariance matrix estimator. *Econometrica*, 60(4), 953–966.
- Granger, C.W.J., Newbold, P. (1974). Spurious regressions in econometrics. *Journal of Econometrics*, 2(2), 111–120.
- Holm, S. (1979). A simple sequentially rejective multiple test procedure. *Scandinavian Journal of Statistics*, 6(2), 65–70.
- Kiefer, N.M., Vogelsang, T.J. (2005). A new asymptotic theory for heteroskedasticity-autocorrelation robust tests. *Econometric Theory*, 21(6), 1130–1164.
- Kiefer, N.M., Vogelsang, T.J., Bunzel, H. (2000). Simple robust testing of regression hypotheses. *Econometrica*, 68(3), 695–714.
- Künsch, H.R. (1989). The jackknife and the bootstrap for general stationary observations. *The Annals of Statistics*, 17(3), 1217–1241.
- Lazarus, E., Lewis, D.J., Stock, J.H., Watson, M.W. (2018). HAR inference: recommendations for practice. *Journal of Business & Economic Statistics*, 36(4), 541–559.
- Newey, W.K., West, K.D. (1987). A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica*, 55(3), 703–708.
- Newey, W.K., West, K.D. (1994). Automatic lag selection in covariance matrix estimation. *The Review of Economic Studies*, 61(4), 631–653.
- Patton, A., Politis, D.N., White, H. (2009). Correction to "Automatic block-length selection for the dependent bootstrap". *Econometric Reviews*, 28(4), 372–375.
- Phillips, P.C.B. (1986). Understanding spurious regressions in econometrics. *Journal of Econometrics*, 33(3), 311–340.
- Politis, D.N., Romano, J.P. (1994). The stationary bootstrap. *Journal of the American Statistical Association*, 89(428), 1303–1313.
- Politis, D.N., White, H. (2004). Automatic block-length selection for the dependent bootstrap. *Econometric Reviews*, 23(1), 53–70.
- White, H. (2000). A reality check for data snooping. *Econometrica*, 68(5), 1097–1126.